

A Needle in a Stack of Magic Cards

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Links. Demo video: TODO · Live app: TODO · Code: github.com/GoldenSpringFox/DataScience-Project/tree/main/Project/EDHCut

Domain

Magic: The Gathering is a trading card game with over 30,000 mechanically unique cards! The most popular way to play is the **Commander** format, which imposes three rules on deckbuilding: exactly **100 different cards**; one **commander** that sits outside the other 99; and every card's colors must be a subset of that commander's *color identity*. Within it, **any card ever printed** is legal (barring ~50 banned).

The consequence of so many options is that players struggle to **cut**. There is always a shiny new card to add, but it is never fun to take a good one out to make space. Making good cuts means knowing which cards are only good *together* - a **synergy package** - and finding those automatically is the problem this project tackles.

Data

We assembled four sources into one SQLite database (185 MB; ~560 MB with derived matrices).

source	what it gives	size
Scryfall	every printed card: cost, type, oracle text, color identity, ~100 fields	31,830 cards, 20 MB
Scryfall Tagger	crowdsourced mechanic labels (themes, tribes, actions) - our only ground-truth-ish account of what a card <i>does</i>	376,800 tags over ~950 types, 55 MB
Archidekt	user-submitted decklists	13,207 decks over 1,001 commanders, 110 MB
EDHREC	aggregate metagame statistics and commander themes	0.5 MB

Tagger labels are deliberately **held out of every clustering step**, which lets us use them later as an independent check rather than an input.

Scraping every decklist would take months, so we sampled: the top 1,000 commanders at ~10 decks each, plus ~1,000 decks on 5 specific commanders, correcting the imbalance with inverse-probability deck weighting. Two biases are unfixable: most players never upload at all, and an uploaded deck is not necessarily one anyone has played.

Card popularity falls off sharply - the median card appears in only 14 decks - so we require at least 3 (18,979 of 31,830 survive), and 16,026 reach the final graph. We also **exclude the commanders themselves**, or every graph would organise itself around the 1,000 commanders rather than the themes inside the decks.

1. Two-card combo - measuring synergy between cards

Problem Description

Which pairs of cards actually belong together? Sharing a deck is weak evidence of synergy: the color-identity rule alone guarantees two white cards meet far more often than a white and a blue one, a precon ships 100 cards together, and *Sol Ring* sits in 82.5% of decks and so correlates with everything. We needed a number that sees through all these problems.

Our Solution

We turned the 13,207 decklists into a graph of cards: each card a node, each surviving pair an edge, and the edge weight our estimate of how strongly the two want to share a deck. Everything in Question 2 is built on this graph, so a lot of effort went here.

We scored every pair with three association metrics, each fixing a flaw in the one before:

1. **Co-occurrence** - how many decks contain both cards. Simple, and dominated by popularity.
2. **Lift** - $(\text{joint} \cdot N) / (\text{count}_A \cdot \text{count}_B)$: how many times more often two cards meet than they would at random. It reads directly - "4× more than chance" - but treats two decks and two hundred as equally convincing.
3. **t-score** - $(\text{observed} - \text{expected}) / \sqrt{\text{observed}}$, which asks not how far above chance a pair sits but how *well-evidenced* that gap is, so a handful of decks cannot produce a headline number.

We also changed what *expected* means. The naive version divides by all ~13,000 decks, quietly assuming any card could have gone into any deck; we divide by the decks *legally able* to run each card. On 20 hand-picked pairs this collapses generic same-color pairs by **72-92%** while real synergy loses a few percent, and leaves colorless pairs untouched - the algebraic check we expected.

Our synergy score is $t\text{-score} \times \log(1 + \text{lift})$. The t-score carries the evidence, the lift carries the effect size, and neither alone was good enough.

A score only ranks pairs, though; it never says which pairings are relationships at all. That decision turned out to be impossible pairwise, so we make it **structurally**: an edge survives only if the two cards keep the same company, measured as the Jaccard overlap of their neighbour sets. The score says *how strong*; the gate says *whether it is real*.

Evaluation

Evaluation Criteria

We evaluated against a hand-labelled keep/drop set of real pairs, using our own domain knowledge: a metric works if it keeps pairs a player would call synergy and drops pairs that merely share a color or a shelf. Generic staples should score low with everything, a card that wants thing A should score high with suppliers of A but not B, and a card almost nobody plays shouldn't score high with anything.

Setup

All three metrics run over the same 13,207 decks, weighted by inverse commander probability, with precon cards down-weighted by how often players actually cut them.

Results & Visualization

Each metric fails in its own direction, and the failures are what built the final formula.

Lift is disqualified outright. Its best pairs all sit at exactly **4,402** - what two cards score when both appear in 3 decks and never apart - while our strongest genuine combo scores 64.

t-score fails in the mirror direction. It kills those coincidences dead, but its own top four is *Sol Ring*, *Arcane Signet* and *Command Tower* in every combination - cards in thousands of decks that go in everything.

The product balances them. Under t-score, 6 of the 9 generic pairs outscore the weakest real pair; multiplying by $\log(1 + \text{lift})$ cuts that to **2 of 9**.

The gate decides what the score cannot. At 0.03 it keeps all 10 synergy pairs and drops 7 of the 9 others. Both survivors are removal spells - *Abrade* + *Lightning Bolt*, *Swords to Plowshares* + *Loran of the Third Path* - which keep the same company without needing each other. That is a different relationship entirely, and what Question 3 takes up.

Generic staples pass every filter, so we caught them by a different property: no above-chance affinity to *anything*. *Sol Ring*'s strongest lift in the format is **1.13** against *Lion Sash*'s **104.6**, and $\text{max lift} < 3.5$ and $\text{play rate} \geq 10\%$ selects exactly 12 cards to exclude.

Impediments

Most of our parameters are mitigations rather than free choices:

problem	what we did
Thin evidence - the median card is in 14 decks	require ≥ 3 decks; below that, a package and a coincidence look identical
Graph too dense to cluster	keep each card's 15 strongest edges, union-symmetrised so a niche card keeps its link to a popular one
Borderline edges	Jaccard gate at ≥ 0.03 ; reading the 0.02–0.03 band by hand found it almost entirely staple noise, 82% of it linking unrelated communities
Pre-constructed decks donate thousands of meaningless pairs	down-weight precon cards by how often players actually cut them

Tools

Algorithms: *lift*, *t-score*, *color-conditioned null model*, *Jaccard neighbourhood overlap*.

Libraries: *pandas*, *numpy*, *scipy*, *SQLite*.

2. "This would go great in my dragon deck!" - recovering card packages from co-occurrence

Problem Description

A good magic deck picks a theme and adds cards from **packages** that serve it. Since you draw only part of your deck each game, redundant synergies matter, and a package promises its cards work together. Can those packages be recovered from co-occurrence alone, by a system that has never read a card? And does a card belong to one package or several?

Our Solution

We ran community detection on the synergy score graph from Question 1 (16,026 cards, 158,294 edges), comparing two algorithms:

1. **Louvain** gives a **hard partition**: every card lands in exactly one community, chosen to maximise modularity. Fast, easy to reason about, and taught in class.
2. **symNMF** gives a **soft** one. It factorises the graph as $S \approx H \cdot H^T$ with $H \geq 0$, so each row of H is a card's membership spread across k topics. A card can belong to three packages with weights rather than being forced into the best one.

To test whether the communities match known themes, we label each with mechanic tags from **Scryfall Tagger**, scoring every tag by $\text{share} \times \log(1 + \text{lift})$ and requiring it to be both common ($\geq 10\%$ of the community) and distinctive ($\text{lift} \geq 8$ over ≥ 4 cards). Either alone gives nonsense: *share* picks tags every card carries, *lift* picks freak tags on four cards. **Those tags never enter the clustering**, so agreement between them and the communities means something.

parameter	value	why
topics k	200	chosen to be comparable to Louvain's community count on the same graph
membership floor	$\text{share} \geq 10\%$, at most 5 topics	below a tenth of a card's mass, membership is noise
k -core floor	$k = 4$	a card must have 4 edges <i>inside</i> a community to belong to it; tested against a hand-labelled set, it removed 11 of 12 cards we had flagged as misplaced and 0 of 12 known core members
seed	42, fixed	symNMF is non-deterministic - see Evaluation Criteria

Evaluation

Evaluation Criteria

The system gets no game knowledge, so success means the packages are recognisable *afterwards*. Three checks: the held-out tags must name each community specifically; color purity must beat same-size random draws; and the result must survive a change of random seed, measured against a **degree-preserving rewiring** of the graph - same cards, same connection counts, structure destroyed.

Setup

symNMF converges in 24 seconds, producing 200 communities against Louvain's 244. A standing guard checks that no oversampled commander quietly becomes a "package".

Results & Visualization

The communities are real and legible: Equipment, Landfall, Extra Turns, Shrines, Cascade, Deserts, Allies and twenty-odd typical packages, all named by tags the clustering never saw.

Our own title case is the best illustration. Dragons come out not as one package but **three**, split by how the deck plays rather than by creature type: an aggressive Rakdos core (79 cards), a multicolor ramp shell (62), and a Grixis control build (41). "Would this go great in my dragon deck?" has three different answers.

The soft/hard difference is the sharpest finding: cards average **1.45** memberships and **41% belong to more than one community**, structure a hard partition cannot represent. *Mondrak*, *Glory Dominus* doubles your tokens, is an expensive staple played in high-power decks, and is a Phyrexian - and the model independently places it 47% in token payoffs, 22% among powerful staples and 21% in Phyrexian typical. One package per thing the card does, without reading a word of it.

That effect concentrates exactly where it matters most - on the popular, multi-purpose cards a player actually agonises over cutting, and the ones a hard partition describes worst.

Neither result is chance. Color purity reaches **56.6%** across the 200 communities against a same-size random null of **19.4%**, and refitting at five seeds reproduces **76%** of the structure against **43%** on the rewired graph.

Impediments

Several methods failed before symNMF worked. **Girvan-Newman** cost ~65 minutes per subgraph, **k-clique percolation** ignores edge weights and collapsed the graph into one 15,479-node blob, and an earlier **NMF** model produced 40 topics that never read as archetypes. **Degree normalisation** improved every metric we could measure while producing visibly worse communities, so we left it off.

One bias survives: controlling for popularity, **31% of three-color cards** land in no community at all, against 3-6% for every other color count - they are legal only in decks covering all three colors, so their partners spread too thin to clear the k-core floor.

Tools

Algorithms: *symNMF*, *Louvain*, *k-core*, *modularity*, *adjusted Rand index*.

Libraries: *networkx*, *scipy*, *numpy*, *ipysigma*.

3. Reading the card explains the card - text similarity vs. co-occurrence

Problem Description

Question 2's model never read a single card. This one reads nothing else. Both produce a list of "cards like this one", so do the lists agree? The answer is a distinction the project needed anyway: suggesting a **replacement** for a card is a different job from suggesting what to build **around** it. Cutting a card raises the first question, keeping it the second.

Our Solution

We built a second similarity measure that never looks at a decklist: every card becomes a vector of what is printed on it, compared by the angle between vectors. One algorithm does the work:

1. **TF-IDF + truncated SVD** - TF-IDF scores a word by how often it appears on *this* card against how many cards use it at all, so a word every card carries counts for almost nothing and a rare one counts for a lot. SVD then compresses the result to 64 numbers that keep the directions the cards actually differ along.

A Magic card looks busy, but holds only five things worth reading, each in the same place on every card ever printed: a **name**, a **mana cost**, a **type line**, a box of **abilities**, and for creatures a **power and toughness**. The rest is decoration, and our model ignores it.

Those parts matter unequally, so each goes into its own block, is scaled to the same length, and is then weighted. Keeping them separate matters: pooled into one bag of words, a single shared type-line word like "Human" outweighs a terse card's actual ability.

block	weight	why
rules text	0.6	what the card actually does, and the only block that describes behaviour
type line	0.15	creature vs artifact vs enchantment separates cards that read alike but play differently
mana cost	0.15	a one-mana and a six-mana version of the same effect are not replacements for each other
power / toughness	0.1	creatures only; redistributed onto the other blocks otherwise

Evaluation

Evaluation Criteria

Success is that the text measure returns cards a player would accept as **replacements**, and that it demonstrably answers a different question from co-occurrence. The first is judged by reading real output - the only honest test with no ground truth - plus a standing check: *Cultivate*'s nearest neighbour must be *Kodama's Reach*, the same card under two names. The second is measurable: we compare **functional categories** against **synergy packages**, using Tagger groups that entered neither model.

Setup

Text vectors cover all 31,623 commander-legal cards - every card, whether or not anyone has ever put it in a deck, which is exactly what co-occurrence cannot do.

Results & Visualization

The text measure finds replacements, and the clearest way to see it is beside the other list.

Both are correct answers to different questions: "what do I put in this slot instead?" wants the first, "is this card doing anything here?" the second.

The two signals are close to unrelated, and that is measurable rather than impressionistic.

Text similarity cannot make that distinction at all - the groups that read most alike are goblins (a package) and counterspells (a category). It also settles two loose ends: *Starfall Invocation* landed in no community because every other boardwipe had already been cut from its neighbourhood, and *Abrade + Lightning Bolt* survived Question 1's gate for the same reason.

One last thing the text space told us is really a fact about the game rather than about our model.

Magic writes in its own narrow dialect, and an English stopword list - which strips "the" and "of" - leaves all of those untouched, so a real share of what our vectors measure is vocabulary the whole game shares rather than anything specific to the card.

Impediments

A **land that taps for white and a creature that taps for white read almost identically**, and early on 14 of *Avacyn's Pilgrim*'s 15 nearest cards were lands. No statistic can discover that a land is never a replacement for a spell - a deck runs ~37 of them regardless - so we encoded that boundary directly. The deeper limit is that text similarity has no idea what a card is *worth*: two cards can read identically and be separated by a decade of power creep, which is why we would not ship replacement suggestions from text alone.

Tools

Algorithms: *TF-IDF*, *truncated SVD*, *cosine similarity*, *t-SNE*.

Libraries: *scikit-learn*, *numpy*, *pandas*.

Future Work

Sharpening what we built. We removed each confounder separately but never measured how much of the raw signal each accounts for, and neighbourhood overlap was simply the first structural statistic we tried. Our 200 packages were chosen to match the hard partition's count and are probably too coarse - a granularity sweep peaked closer to 1,200 - and neither the gate nor the membership floor knows anything about color legality, which is why 31% of three-color cards go unassigned.

Question 3 - reading cards. Figure 19 is the clearest lead: our model treats rules text as English when it plainly is not, so the next version should be built for Magic specifically, with its own stopword list and a decomposition fitted to this dialect. We would also predict a card's package from its rules text alone, turning Question 2's output into labels and Question 3's vectors into features.

The tool this was all for. The knowledge base exists to power a cut recommender: score each of a deck's 99 cards by how weakly it attaches to that deck's packages, and suggest the loosest. Ground truth already exists in our data - thousands of our decks began as a preconstructed product, so we can see **which cards real players actually cut** when upgrading one, and check whether our score agrees.

Conclusion

Co-occurrence between cards is mostly artifact - legality rules, the products people buy, and the handful of cards that go in everything explain most of it - and no statistic computed from a pair in isolation separates the real relationships from the rest. What worked was to stop asking about the pair and start asking about the company each card keeps.

What survives that filtering is structural. A system given no card text, no types and no rules of the game recovered goblins, landfall, equipment and three kinds of dragon deck, and we could name them with labels it had never seen. Reading the cards themselves answered a different question entirely - which card could *replace* this one - and a real cut recommender needs both. The knowledge base is built and validated; the tool that would consume it is not, and what we have is a solid baseline for it.

Afterword

It was incredibly fun to work on this project. As someone who is both visually oriented and constantly thinking about this hobby, it's super cool seeing these connections i've been developing in my head take form on these graphs. It's incredible how simple and accessible exploring the data is these days, and i'm definitely intending to finish this tool and uploading it to the internet for other people to use. There is a widespread epidemic of EDHREC-ification, where deckbuilding data isn't accessible enough, so people go to edhrec and put in only genericly popular cards, instead of looking for cool niche synergy pieces.

By the way, we found the needle! better drop it back for the next group to find: